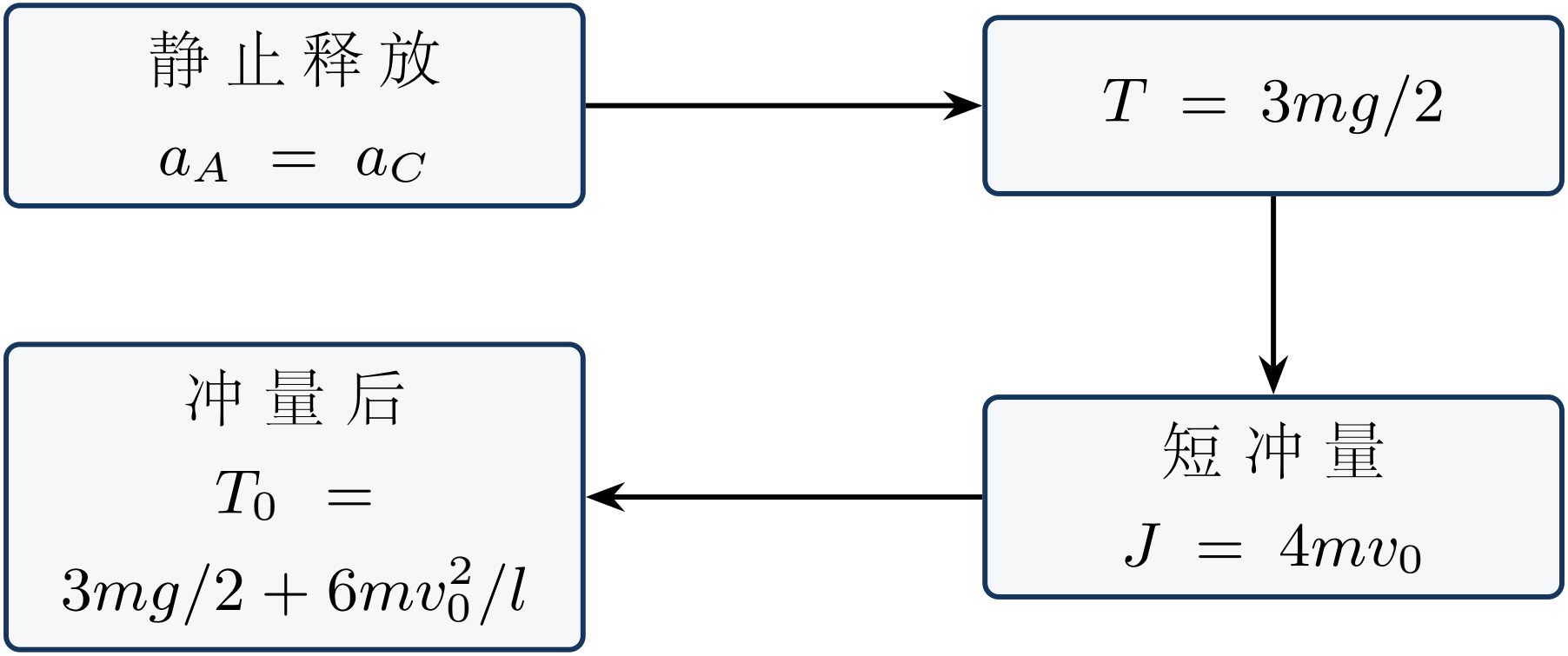

广义等效质量 $4m$；冲量后总动能 $2mv_0^2$。